

UM10492

PTN3460 eDP to LVDS Bridge IC Application Board User's Manual

Rev. 0.2 - 6/29/11

Document information

Info	Content
Keywords	PTN3460, DisplayPort, eDP, LVDS, bridge, application board
Abstract	This user manual presents demonstration / application board capability of interfacing an (embedded) DisplayPort source to a LVDS panel. The application board (nicknamed "DPLVDS1") is intended for use as an evaluation and customer demonstration tool, as well as a reference design.

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Revision history

Rev	Date	Description
0.1	03012011	Drafted by Linda Lu
0.2	06152011	Drafted by Linda Lu

1. Introduction

This document describes the user manual of PTN3460 AUO application board, including

- Overall PCB connectors, jumpers, and power supplies
- Equipments/Tools that this board will be interfacing with during bench testing
- System level connections such as cables and connectors that this board will be plugged into

This application board is intended to demonstrate the bridging capabilities of PTN3460 on DP to LVDS conversion. It can also be used to evaluate competitor's ICs such as Parade's PS8615, which is pin-to-pin compatible with PTN3460.

A separate document "application test plan" will be provided to list items to be verified on the system level.

1.1 Purposes

- For internal engineers to evaluate the performance of PTN3460 and to develop firmware, including collecting and verifying system level features/performances/functionalities such as
 - o Verify power management schemes
 - o Power sequence
 - o Power consumption measurement during various operating modes.
 - o Allow access to test points and jumpers for measurement and configuration purposes
 - o Flash over AUX
 - o Programming test via I2C
- For marketing to demonstrate DP-LVDS to customers in the field
 - o Functional and interoperability test.
 - o This board should be connected to a DP or an eDP source.
 - o This board is powered by ATX 20-pin power supply, or
 - o External power supplies with +3.3V (1A), +5V (1A), +12V (2A)
 - o This board is connected directly to AUO 21" LVDS panel, or
 - o Connect to NB 14" LVDS panel via an AUO-NB adaptor board.
- For customers to evaluate PTN3460
 - o Use I²C to program EDID.

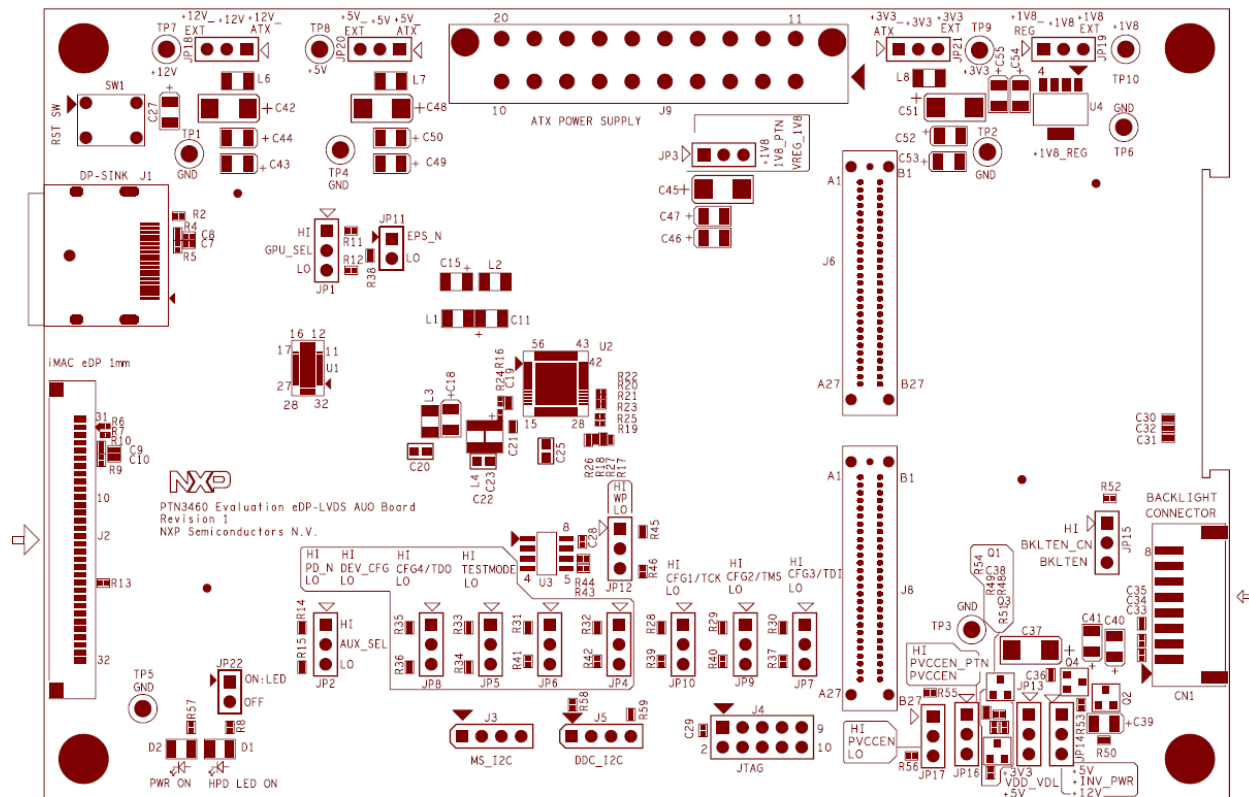
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2. General Description

2.1 Top and Bottom assembly drawings of the PTN3460 application board is shown below.

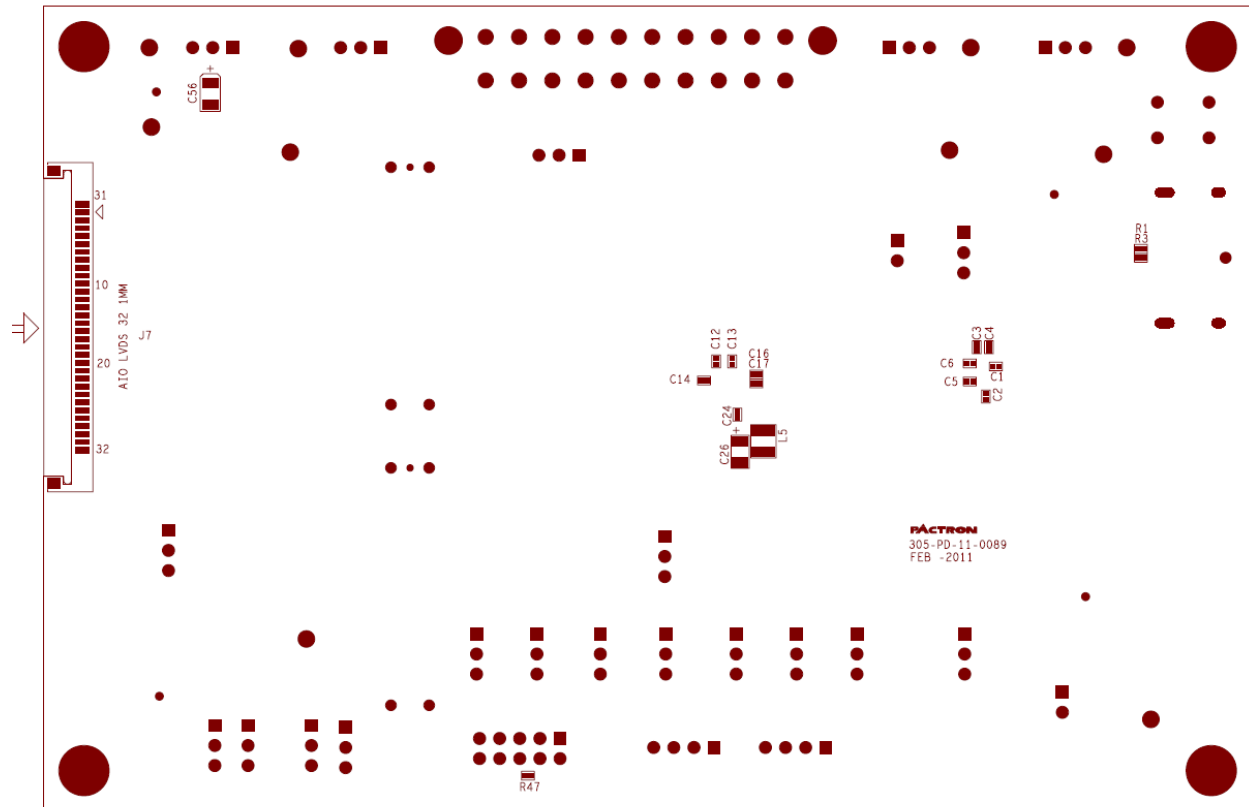


TOP ASSEMBLY

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BOTTOM ASSEMBLY

Figure 1 PTN3460 Application Board Top & Bottom Assembly Drawings

The PTN3460 AUO Application board has the following features:

- Jumpers select multiplex between DP and eDP (iMac) sources.
- Stuff options for PT3460 different pin configurations.
- Stuff options for PS8615 test.
- Four (4) groups of jumpers for pin configuration.
- Other jumpers for test options.
- One Serial EEPROM to store EDID Read/Write via I2C.
- One (1) I²C header bringing out I²C pins (SCL, SDA, GND) to interface with a I²C-Bird dongle to program S-EEPROM or to read iMac System Bus.
- One (1) I2C header for DDC control.
- One (1) JTAG for FW download.
- One (1) ATX 20-pin power supply, with 12V/5V/3.3V DC inputs.
- External power supplies 12V (2A), 5V (1A), 3.3V (1A).
- Fix regulator to provide 1.8V±10% power supply.
- Two (2) soft touch connectors (not loaded) for Agilent differential probes to capture LVDS signals.
- One 30-position 1mm connector with cables to connect to AUO 21" LVDS panel.
- One 1x8 connector with cable to connect to Back Light Inverter.

3. Hardware Requirements

1. Item 1 – iMac 27" AIO, eDP source with 30 pos eDP cable
2. Item 2 – VTG5225-DP, DP source with DP 1.1 cable
3. Item 3 – DPA-400, AUX analyzer, via iMAC eDP-DP adaptor with 2 DP 1.1 cables and 2 30 pos eDP cables
4. Item 4 – UFG-04, LVDS grabber, via AUO-grabber adaptor with 2 MDR-26 twisted cables
5. Item 5 – Desk top PC with PCIe slot to install LVDS grabber
6. Item 6 – AUO 21" LVDS panel with 30 pos LVDS cable
7. Item 7 – FS2 with 2x5 JTAG connection for FW download
8. Item 8 – I2C Bird with 1x4 header connection for s-EEPROM R/W

4. Board Specifications

4.1 General Description

- Layers: 4 layers expected – trace, ground, VCC, trace
- Size: 4"x6"
- Material: FR4
- Thickness: 62 mil
- Impedance: 50ohm single-end, 100ohm differential on DP and LVDS signal pairs.

5. Connector Specifications

5.1 Connectors

J1	DP CONN SINK	conn_47272-0001	Molex	47272-0001
J2	FI-X30SSL-HF	conn_FI-X30SSL-HF	JAE	FI-X30SSL-HF
J3,J5	HEADER 4	hdr_4x1	Sullins	PBC04SAAN
J4	HEADER, 2x5	hdr_5x2	Sullins	PBC05DAAN
J6,J8	SOFT TOUCH CONN E5387-68701	conn_E5387-68701	Agilent	E5387-68701
J7	FI-XB30SRL-HF11	conn_FI-XB30SRL- HF11	JAE	FI-XB30SRL-HF11
J9	MOLEX ATX PWR PN 39- 29-9202	cn_molex_minifit20 p_vt	Molex	39-29-9202
CN1	S8B-PH-SM4-TB(LF)(SN)	conn_8x1_2mm	JST Sales	S8B-PH-SM4-TB(LF)(SN)

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5.2 Cables

	Test Cable Location	Test Cables		
1	J1	DP 1.1 cable	Purchase ready made cable	
2	J2	iMac 30 pos eDP 1mm cable	Made from LVDS 30 pos cable kit	
3	J3, J5	I2C-Bird 1x4 cable	Ready made with I2C-Bird box	
4	J4	JTAG 2x5 ribbon cable	Ready made with FS2 box	
5	J6, J8	Agilent soft-touch probes	90 pin differential probe E5387A (2)	
6	J7	AUO 30 pos LVDS 1mm cable	Made from LVDS 30 pos cable kit	
7	CN1	Back light inverter 1x8	Made from accessories	

5.3 Jumpers

Jumper #	Signal Names	Jumper Settings	Default Setting
JP1	U1 MUX GPU_SEL	1-2 HIGH, select DP inputs 2-3 LOW, select eDP inputs	1-2
JP2	U1 MUX AUX_SEL	1-2 HIGH, select AUX from DP inputs 2-3 LOW, select AUX from eDP inputs	1-2

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JP3	1V8_PTN	1-2, 1.8V from on board regulator 2-3, 1.8V from Buck converter	1-2
JP4	TESTMODE	1-2 HIGH, CFG[4:1] = JTAG PINS 2-3 LOW, CFG[4:1] = CONFIG PINS	2-3
JP5	DEV_CFG	1-2 HIGH, I2C Bus Master OPEN, I2C Bus Slave (0C0h) 2-3 LOW, I2C Bus Slave (040h)	2-3
JP6	CFG4/TDO	1-2 HIGH, LVDS output swing 400 mV OPEN, LVDS output swing 300 mV 2-3 LOW, LVDS output swing 250 mV	1-2
JP7	CFG3/TDI	1-2 HIGH, LVDS CLK FRQ 0.5% OPEN, LVDS CLK FRQ 1% 2-3 LOW, LVDS CLK FRQ 0%	1-2
JP8	PD_N	1-2 HIGH, Operation mode 2-3 LOW, Force power down	1-2
JP9	CFG2/TMS	1-2 HIGH, JEIDA or VESA format (18 bpp) OPEN, JEIDA format (24 bpp) 2-3 LOW, VESA format (24 bpp)	1-2
JP10	CFG1/TCK	1-2 HIGH, Dual LVDS bus 2-3 LOW, Single LVDS bus (note: does not work, use FW control)	1-2
JP11	EPS_N	ON, Use external 3/3V/1.8V option OFF, Use internal 1.8V LDO	1-2
JP12	WP	1-2 HIGH, WP for S-EEPROM	2-3

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		2-3 LOW, No WP S-EEPROM	
JP13	VDD_VDL	1-2, select +3V3 for Panel 2-3, select +5V for Panel	1-2
JP14	+INV_PWR	1-2, select +5V for Back light inverter 2-3, select +12V for Back light inverter	1-2
JP15	BKLTEN_CN	1-2 HIGH, Back Light Enable is ON always 2-3 BKLTEN, Control by Firmware	1-2
JP16	PVCCEN_PTN	1-2 HIGH, PVCCEN is ON always 2-3 PVCCEN, Control by firmware	1-2
JP17 (for PS8615 only)	PVCCEN (pin 33 is used as I2C_Addr/GPIO in PS8615)	1-2 HIGH, I2C_ADDR 0xB0h-0xBFh 2-3 LOW, I2C_ADDR 0x10h_0x1Fh	Not loaded
JP18	+12V	1-2, select from ATX power supply 2-3, select from external power supply	2-3
JP19	+1V8	1-2, select from U4 regulator 2-3, select from external power supply	1-2
JP20	+5V	1-2, select from ATX power supply 2-3, select from external power supply	2-3
JP21	+3V3	1-2, select from ATX power supply 2-3, select from external power supply	2-3
JP22	AIOC_HPDP	ON: HPD drives Green LED OFF: No LED drive to measure power	1-2

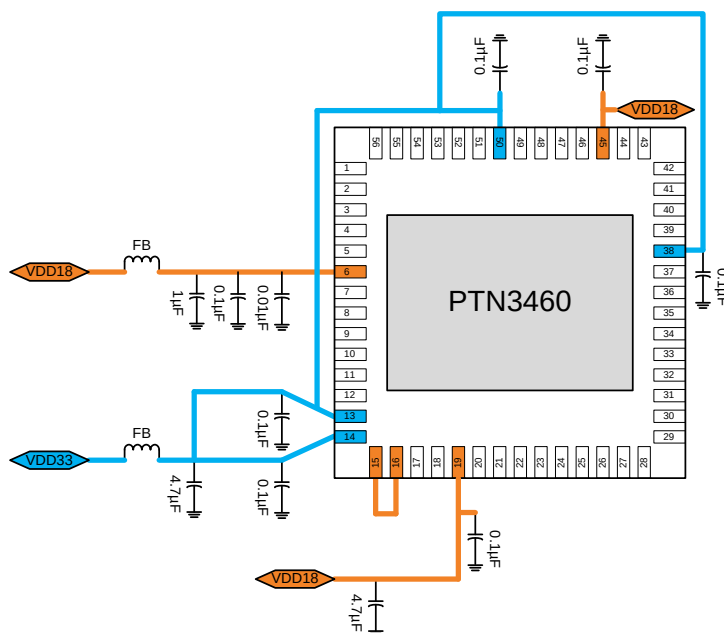
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6. Stuffing Options

6.1 PTN3460 NXP Pinning



6.2 Stuffing Locations

Location	Function/Value	PTN3460 (default pinning)
R16	Join +3V3_IO with +3V3_LDO	No Load
R17	Pin 27 = CFG4/TDO	Load
R18	Pin 26 = BKLTEN_PTIN	Load
R19	Pin 28 = PWMO_PTIN	Load
R20	Pin 32 = LVSDE_N_PTIN	Load

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R21	Pin 31 = LVSDE_P_PTN	Load
R22	Pin 32 = PWMO	No Load
R23	Pin 31 = BKL TEN	No Load
R24	+3V3_LDO for pin 13, 14	Load
R25	Pin 28 = GND	No Load
R26	Pin 26 = GND	No Load
R27	Pin 27 = GND	No Load
C27	1uF for RST_N line	No Load
C22	0.47uF	No Load
C23	4.7uF	No Load
L4	FB	No Load
C25	0.47uF	No Load
C26	4.7uF	No Load
L5	10uH	No Load
C15	2.2uF	Load

7. References

1. Data Specification, PTN3460_DP_LVDS_Bridge_v1.7.pdf
2. Schematic, DP-LVDS-AUO rev1.14 rew.pdf
3. BOM, DP-LVDS-AUO rev1.14.xls
4. PTN3460 PCB Layout Guideline.pdf
5. Allegro layout, PTN3460 Evaluation eDP-LVDS AUO Board_PCB_0089_021811-1.brd
6. Test Cables, PTN3460 Test Cables.pdf

PTN3460 Eval. Board and iMAC eDP-to-LVDS Bridge to AUO Panel Rev.1.14 with Reworks

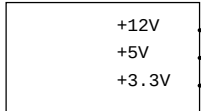
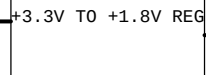
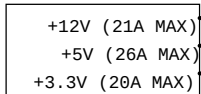
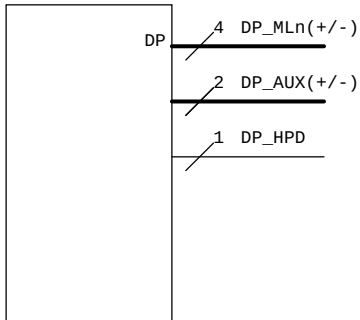
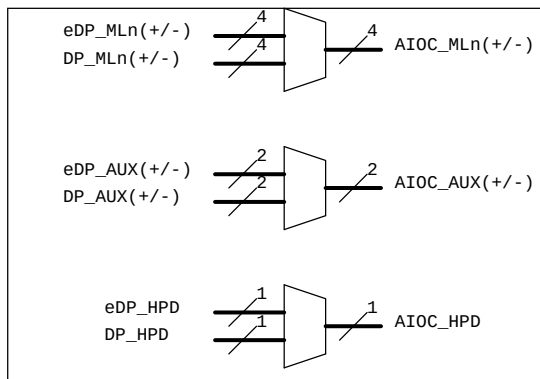
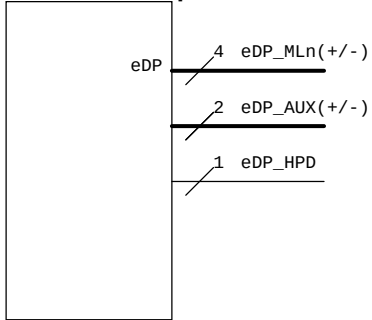
REVISION HISTORY :

- 0.1 Nov.11 2010
 - DP-LVDS starts based on DPVGA4
- 1.1 Dec.20 2010
 - Design Review
- 1.2 Dec.21 2010
 - PTN3460 Power Review
- 1.3 Dec. 22, 2010
 - BOM
- 1.4 Dec. 23, 2010
 - Change JP8 to HDR3
 - Separate two LVDS connetors into two board
- 1.5 Jan. 03, 2011
 - Swap LVDSDE_P and LVDSDE_N signal names
- 1.6 Jan. 10, 2011
 - Separate +3V3_I0 from +3V3_LD0
- 1.7 Jan. 13, 2011
 - Add PS8615 mounting capability
 - Re-number reference designation
- 1.8 Jan. 17, 2011
 - Update placement note on page 6
 - J3, J5 uses I2CBIRD pinout
- 1.9 Jan. 18, 2011
 - Clean up netlist
 - Change 0 Ohm to 0201 size
- 1.10 Jan. 27, 2011
 - Change some 0.1uF to 0.01uF
 - Flip JP13, JP15
- 1.11 Feb. 04, 2011
 - Swap U1 MUX input 1, 2 for better routing
- 1.12 Feb. 07, 2011
 - More Swap U1 to improve routing
 - Re-assign J6, J8
- 1.13 Feb. 11, 2011
 - Remove page references on U2-31, 32
 - Re-arrange some pins on J6, J8
 - Flip JP13, JP15 back
- 1.14 Feb. 15, 2011
 - Change PWR test pins
 - Add silk screen labels
- 1.14 re-work
 - Add 10K pull up R60 to U4-1

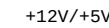
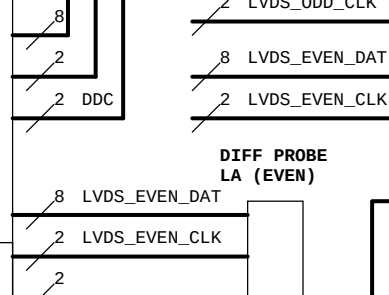
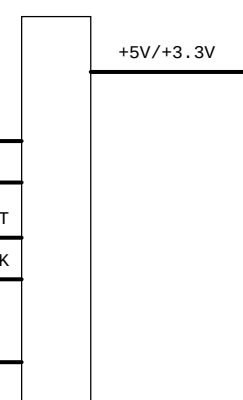
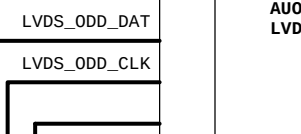
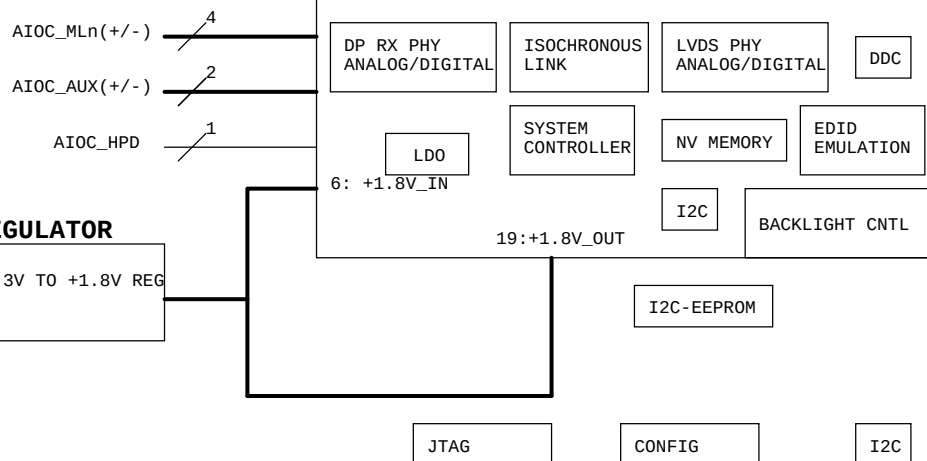
Page# Page Title

- 1 Index
- 2 Main Board Diagram
- 3 iMAC eDP, DP Mux
- 4 PTN3460
- 5 Config Jumper, JTAG
- 6 LA, AUO LVDS, Back Light Inverter Connectors
- 7 ATX Switching Power Supply

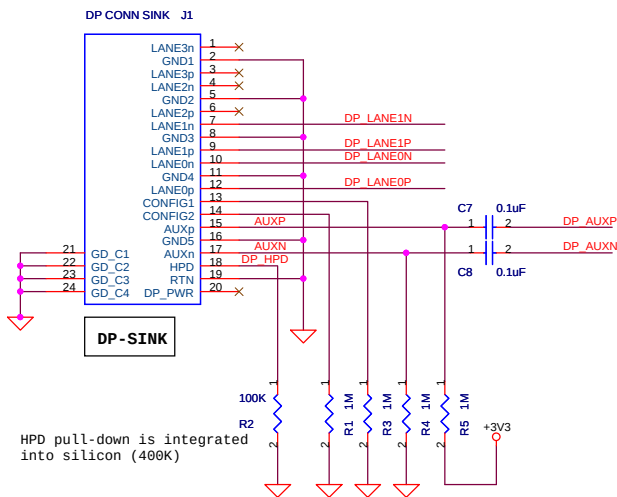
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Title INDEX		
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Size B	Document Number	Rev REW
Date: Friday, June 17, 2011		Sheet 1 of 7



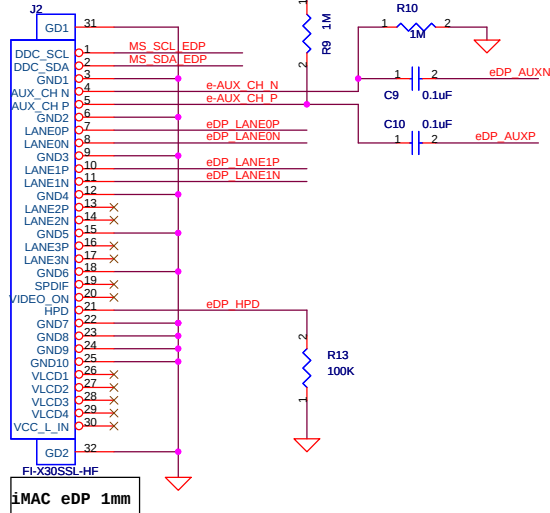
NO LOAD LIST FOR PTN3460:
C22, C23, C25, C27, C29
JP17, J6, J8
L4, L5
R16, R22, R23, R25, R26, R27



NXP Semiconductors			
Title			
Main Board Block Diagram			
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	Rev new		
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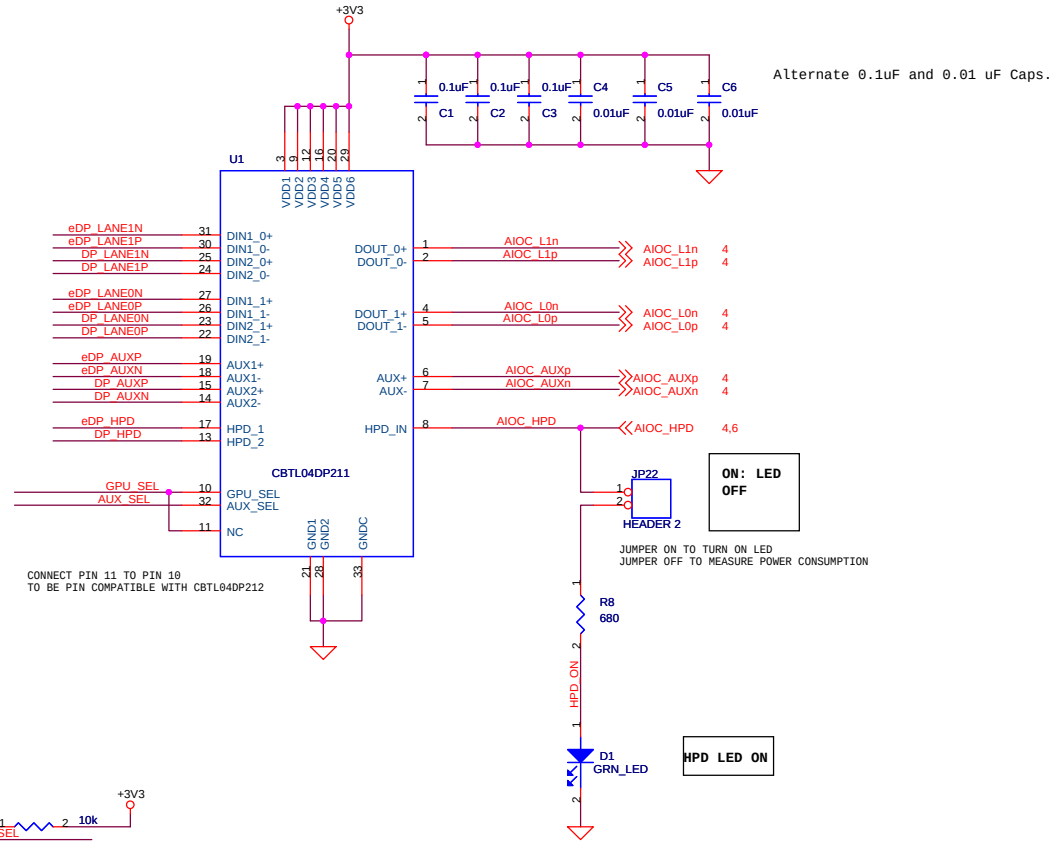
PINOUT FROM APPLE IMAC K74



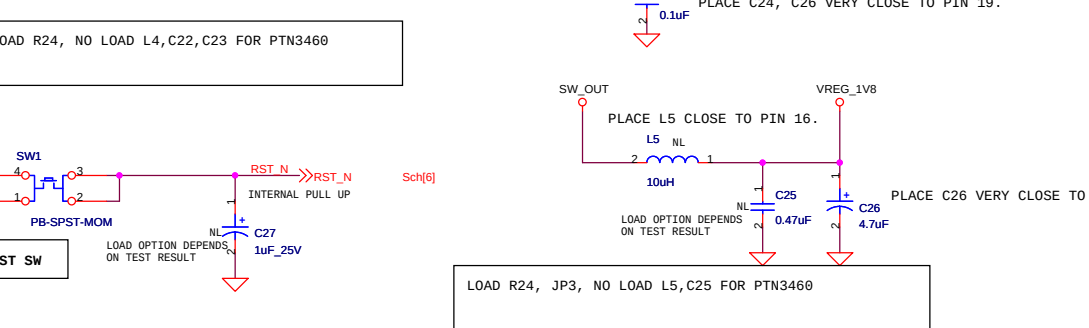
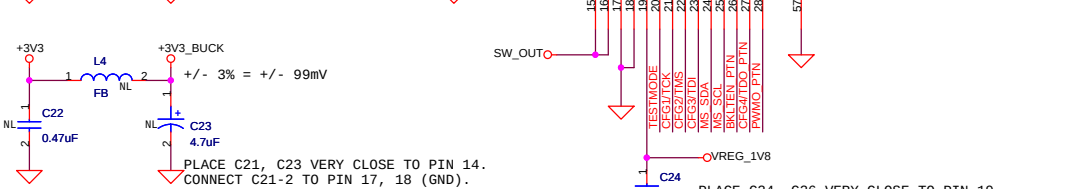
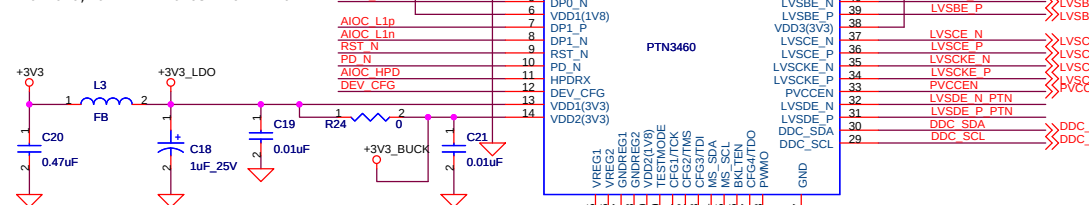
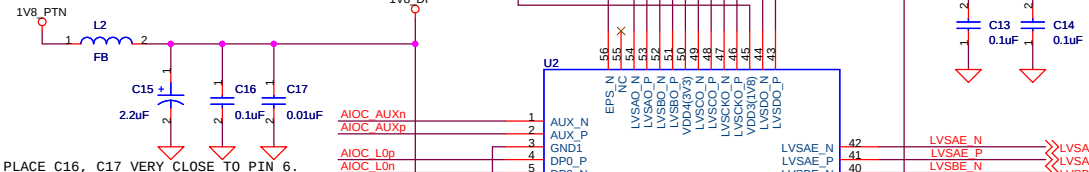
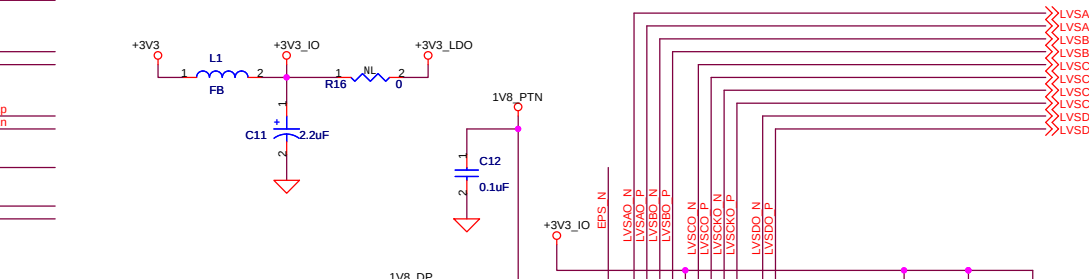
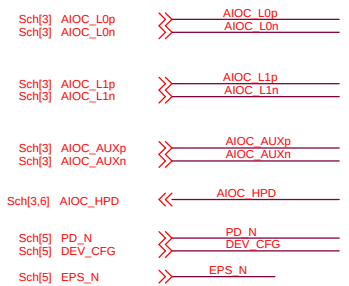
JUMPER 1-2, HIGH, DIN2, HPD2 (DP)
JUMPER 2-3, LOW, DIN1, HPD1 (eDP)



JUMPER 1-2, HIGH, AUX2 (DP)
JUMPER 2-3, LOW, AUX1 (eDP)

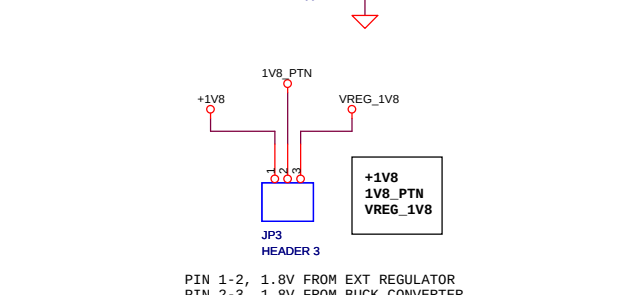
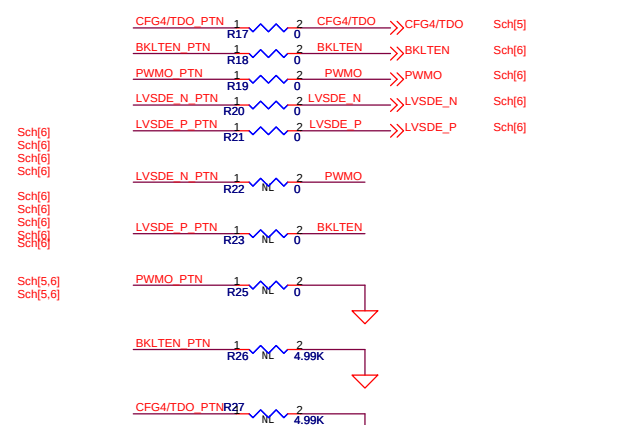


Title		
eDP, DP SINK MUX		
Size	Document Number	Rev 1.12
Date:	Friday, June 17, 2011	Sheet 3 of 7

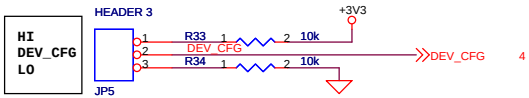


PTN3460 PACKAGE OPTION:
RE-CONFIG PIN 12,26,27,28,31,32,33,56

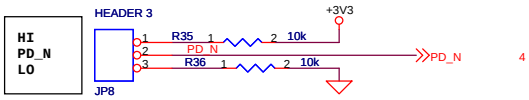
FOR PTN3460 (NXP): INSTALL R17-R21 - DO NOT
INSTALL R22-R27.



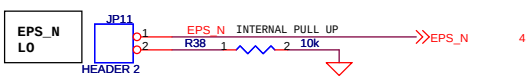
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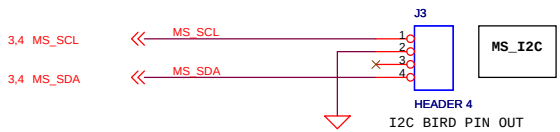
JUMPER 1-2, HIGH, I2C BUS MASTER (READING FROM EXTERNAL EEPROM)
OPEN: I2C BUS SLAVE, HIGH ADDRESS (000-BFH)
JUMPER 2-3, LOW, I2C BUS SLAVE(010-1FH)



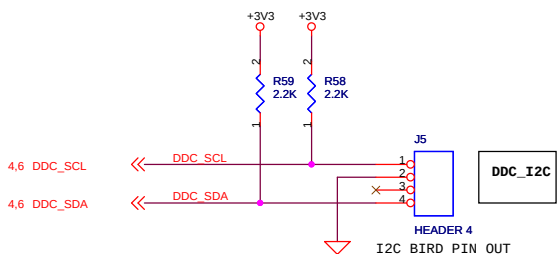
JUMPER 2-3, FORCE POWER DOWN
JUMPER 1-2, PD_N HIGH TO OPERATE



JUMPER ON TO USE EXT 3.3V/1.8V OPTION;
JUMPER OFF TO USE INT 1.8V LDO.

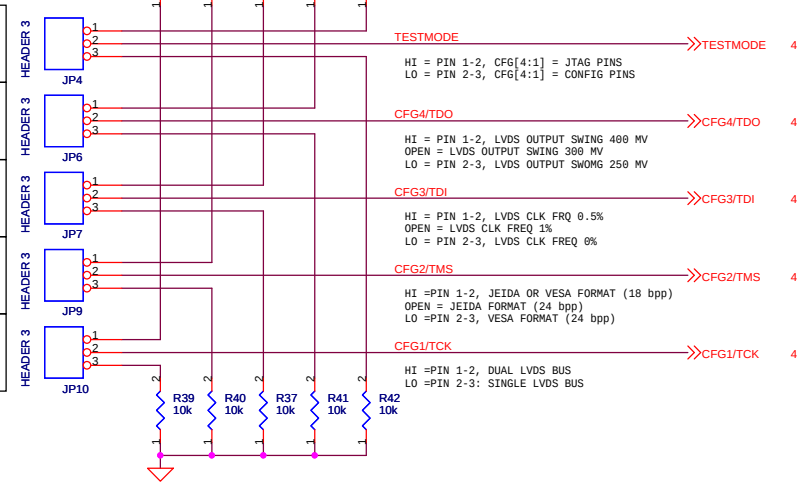


I2C BIRD PIN OUT



I2C BIRD PIN OUT

HI TESTMODE LO
HI CFG4/TDO LO
HI CFG3/TDI LO
HI CFG2/TMS LO
HI CFG1/TCK LO



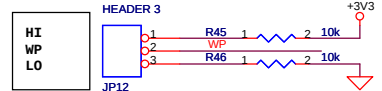
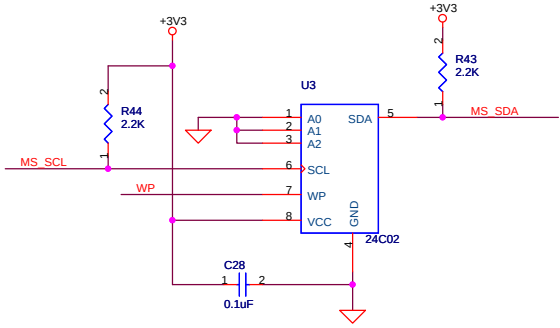
TESTMODE
HI = PIN 1-2, CFG[4:1] = JTAG PINS
LO = PIN 2-3, CFG[4:1] = CONFIG PINS

CFG4/TDO
HI = PIN 1-2, LVDS OUTPUT SWING 400 MV
OPEN = LVDS OUTPUT SWING 300 MV
LO = PIN 2-3, LVDS OUTPUT SWING 250 MV

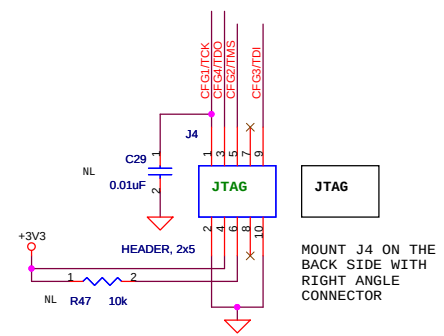
CFG3/TDI
HI = PIN 1-2, LVDS CLK FRQ 0.5%
OPEN = LVDS CLK FREQ 1%
LO = PIN 2-3, LVDS CLK FREQ 0%

CFG2/TMS
HI = PIN 1-2, JEIDA OR VESA FORMAT (18 bpp)
OPEN = JEIDA FORMAT (24 bpp)
LO = PIN 2-3, VESA FORMAT (24 bpp)

CFG1/TCK
HI = PIN 1-2, DUAL LVDS BUS
LO = PIN 2-3, SINGLE LVDS BUS



JUMPER 1-2, HIGH, WP
JUMPER 2-3, LOW, NO WP

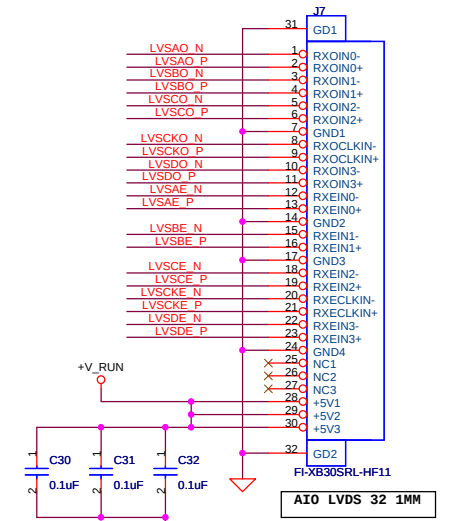
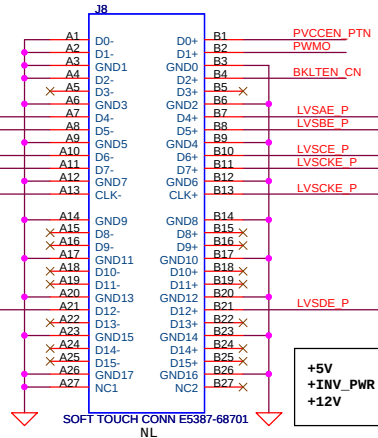
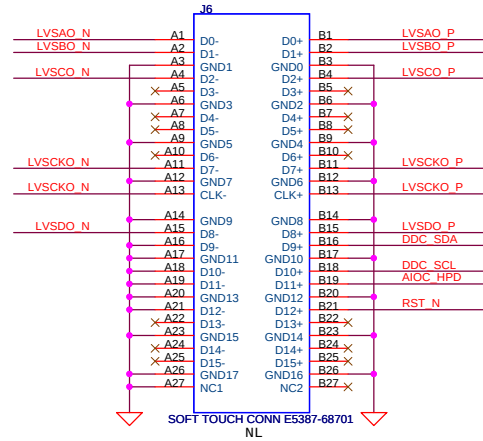
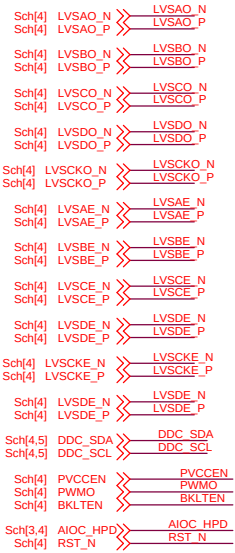


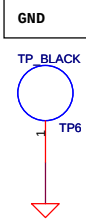
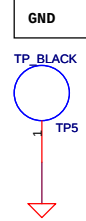
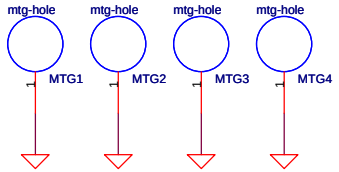
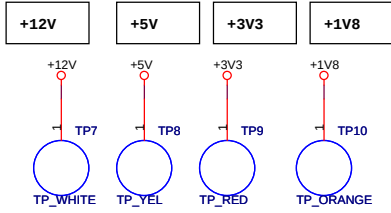
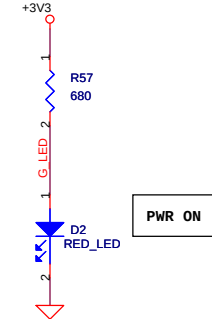
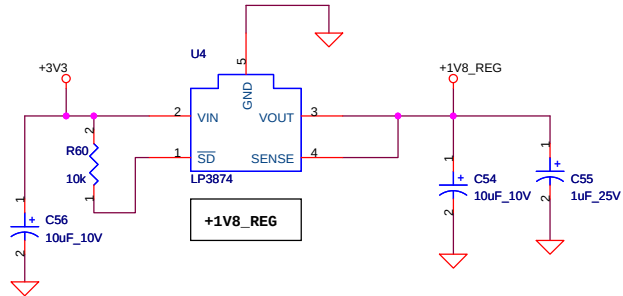
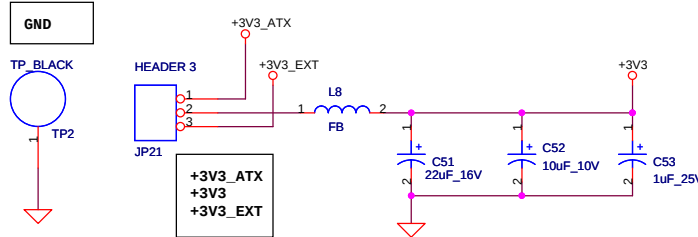
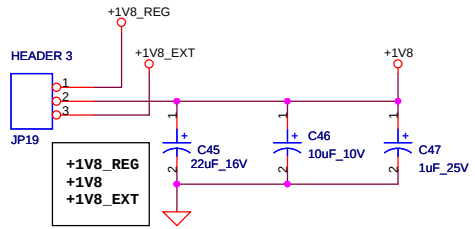
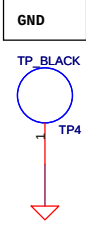
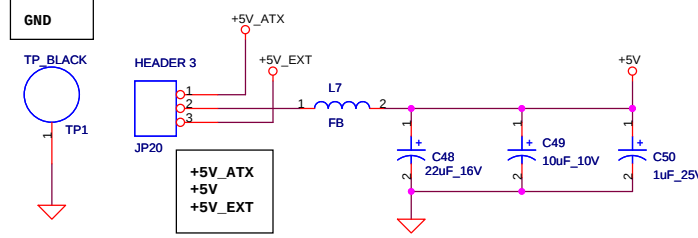
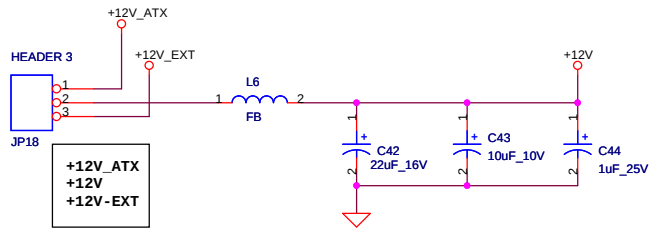
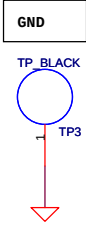
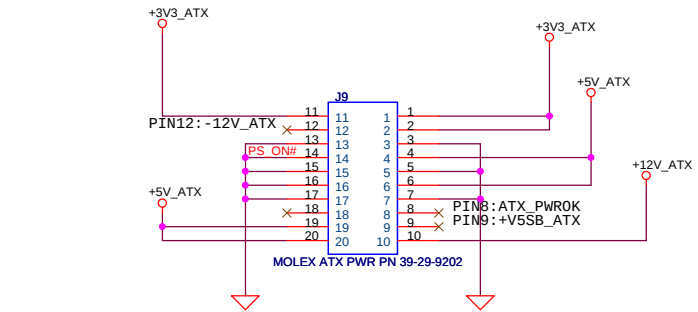
MOUNT J4 ON THE BACK SIDE WITH RIGHT ANGLE CONNECTOR

Title		
PTN3460 AND JTAG		
Size	Document Number	Rev
Date:	Friday, June 17, 2011	Sheet 5 of 7

TO BE USED WITH 90-PIN DIFFERENTIAL LOGIC PROBE E5387A

LVDS DUAL BUS 24bpp
(8-BITS) AUO PANEL





Title		
ATX SWITCHING POWER SUPPLY		
Size	Document Number	Rev
B	Add 10K from U4-1 to U4-2	1.14
Date:	Friday, June 17, 2011	Sheet 7 of 7

INDEX Revised: Thursday, January 27, 2011

eDP-LVDS AUO Revision: 1.10

NXP Semiconductors

1251 McKay Dr. M/S-60SJ, San Jose, CA 95131-1706

Tel: (408)-474-7448, Fax: (408)-474-8334

Bill Of Materials January 28,2011 10:21:32

Item	QTY	Reference	Value	PCB Footprint	Tolerance	Manufacturer	Manufacturer Number	Distributor	Distributor Number	Load Option
1	1	CN1	S8B-PH-SM4-	conn_8x1_2m m		JST Sales	S8B-PH-SM4-TB(LF)(SN)	Digikey	455-1755-2-ND	LOAD
2	20	C1,C2,C3,C7,C8,C9,C10, C12,C13,C14,C16,C24,C28, C30,C31,C32,C33,C34,C35, C36	0.1uF	rc0402	10%	Yageo	CC0402KRX5R6B B104	Digikey	311-1336-2-ND	LOAD
3	7	C4,C5,C6,C17,C19,C21,C38	0.01uF	rc0402	10%	TAIYO	UMK105B7102K V-F	Digikey	587-1220-2-ND	LOAD
4	2	C11,C15	2.2uF	rc1206	10%	Murata	GRM31MR71A2 25KA01L	Digikey	490-1803-2-ND	LOAD
5	9	C18,C39,C40,C41,C44,C47, C50,C53,C55	1uF_25V	eia_3216	20%	Nichicon	F931E105MAA	Digikey	493-2387-2-ND	LOAD
6	1	C20	0.47uF	rc0603	10%	Murata	GRM188R71C47 4KA88D	Digikey	490-3295-2-ND	LOAD
7	2	C22,C25	0.47uF	rc0603	10%	Murata	GRM188R71C47 4KA88D	Digikey	490-3295-2-ND	NO LOAD
8	1	C23	4.7uF	rc1206	10%	TDK	C3216X7R1C475 K/1.60	Digikey	445-1385-2-ND	NO LOAD
9	1	C26	4.7uF	rc1206	10%	TDK	C3216X7R1C475 K/1.60	Digikey	445-1385-2-ND	LOAD

10	1	C27	1uF_25V	eia_3216	20%	Nichicon	F931E105MAA	Digikey	493-2387-2-ND	NO LOAD
11	1	C29	0.01uF	rc0402	10%	TAIYO	UMK105B7102K V-F	Digikey	587-1220-2-ND	NO LOAD
12	5	C37,C42,C45,C48,C51	22uF_16V	eia_6032	20%	AVX	TAJC226M016R NJ	Digikey	478-3901-2-ND	LOAD
13	6	C43,C46,C49,C52,C54,C56	10uF_10V	eia_3216	10%	Kemet	T491A106K010A T	Digikey	399-3684-2-ND	LOAD
14	1	D1	GRN_LED	led-ln1351c		Panasonic	LN1351C	Digikey	P501TR-ND	LOAD
15	1	D2	RED_LED	led-ln1251c		Panasonic	LN1251C	Digikey	P500TR-ND	LOAD
16	19	JP1,JP2,JP3,JP4,JP5,JP6,JP7,JP8,JP9,JP10,JP12,JP13,JP14,JP15,JP16,JP18,JP19,JP20,JP21	HEADER 3	hdr_3x1		Sullins	PBC03SAAN	Digikey	S1011E-03-ND	LOAD
17	2	JP11,JP22	HEADER 2	hdr_2x1		Sullins	PBC02SAAN	Digikey	S1011E-02-ND	LOAD
18	1	JP17	HEADER 3	hdr_3x1		Sullins	PBC03SAAN	Digikey	S1011E-03-ND	NO LOAD
19	1	J1	DP CONN SINK	conn_47272-0001		Molex	47272-0001	Mouser	47272-0001	LOAD
20	1	J2	FI-X30SSL-HF	conn_FI-X30SSL-HF		JAE	FI-X30SSL-HF	Mouser	FI-X30SSL-HF	LOAD
21	2	J3,J5	HEADER 4	hdr_4x1		Sullins	PBC04SAAN	Digikey	S1011E-04-ND	LOAD
22	1	J4	HEADER, 2x5	hdr_5x2		Sullins	PBC05DAAN	Digikey	S2011E-05-ND	LOAD
23	2	J6,J8	SOFT TOUCH CONN	conn_E5387-68701		Agilent	E5387-68701	Agilent	E5387-68701	NO LOAD
24	1	J7	FI-XB30SRL-HF11	conn_FI-XB30SRL-HF11		JAE	FI-XB30SRL-HF11	Mouser	FI-XB30SRL-HF11	LOAD

25	1	J9	MOLEX ATX PWR PN 39- 29-9202	cn_molex_minf it20p_vt		Molex	39-29-9202	Mouser	538-39-29- 9202	LOAD
26	6	L1,L2,L3,L6,L7,L8	FB	rc1206		Laird-Signal	HI1206N101R- 10	Digikey	240-2408-2- ND	LOAD
27	1	L4	FB	rc1206		Laird-Signal	HI1206N101R- 10	Digikey	240-2408-2- ND	NO LOAD
28	1	L5	10uH	rc1210	10%	Panasonic	ELJ-PA100KF2	Digikey	PCD2343TR- ND	NO LOAD
29	2	Q1,Q2	MOSFET P- CH	SOT23-3		Vishay	SI2307CDS-T1- GE3	Digikey	SI2307CDS- T1-GE3TR-	LOAD
30	2	Q3,Q4	MOSFET N- CH	SOT23-3		Diodes	BSS138-7-F	Digikey	BSS138- FDITR-ND	LOAD
31	7	R1,R3,R4,R5,R9,R10,R48	1M	rc0402	5%	Panasonic	ERJ-2GEJ105X	Digikey	P1.0MJCT-	LOAD
32	5	R2,R13,R49,R50,R51	100K	rc0402	5%	Stackpole	RMCF0402JT100 K	Digikey	RMCF0402J T100KTR-ND	LOAD
33	8	R6,R7,R17,R18,R19,R20, R21,R24	0	rc0201	Jumper	Panasonic	ERJ-1GE0R00C	Digikey	P0.0AGTR-	LOAD
34	2	R8,R57	680	rc0402	5%	Stackpole	RMCF0402JT680 R	Digikey	RMCF0402J T680RTR-	LOAD
35	26	R11,R12,R14,R15,R28,R29, R30,R31,R32,R33,R34,R35, R36,R37,R38,R39,R40,R41, R42,R45,R46,R47,R52,R54, R55,R56	10k	rc0402	5%	Stackpole	RMCF0402JT10K 0	Digikey	RMCF0402J T10K0TR-ND	LOAD
36	4	R16,R22,R23,R25	0	rc0201	Jumper	Panasonic	ERJ-1GE0R00C	Digikey	P0.0AGTR- ND	NO LOAD
37	2	R26,R27	4.99K	rc0402	1%	Stackpole	RMCF0402FT4K 99	Digikey	RMCF0402F T4K99TR-ND	NO LOAD
38	4	R43,R44,R58,R59	2.2K	rc0402	5%	Yageo	RC0402JR- 072K2L	Digikey	311- 2.2KJRTR-	LOAD
39	1	R53	47k	rc0402	5%	Panasonic	ERJ-2GEJ103X	Digikey	P10KJCT-ND	LOAD

40	1	SW1	PB-SPST-MOM	SW_PB_CKN		C&K	PTS645SK43SMT RLFS	Digikey	CKN9084TR-ND	LOAD
41	6	TP1,TP2,TP3,TP4,TP5,TP6	TP_BLACK	term_125_065		Keystone	5011	Digikey	5011K-ND	LOAD
42	1	TP7	TP_WHITE	term_125_065		Keystone	5012	Digikey	5012K-ND	LOAD
43	1	TP8	TP_YEL	term_125_065		Keystone	5014	Digikey	5014K-ND	LOAD
44	1	TP9	TP_RED	term_125_065		Keystone	5010	Digikey	5010K-ND	LOAD
45	1	TP10	TP_ORANGE	term_125_065		Keystone	5013	Digikey	5013K-ND	LOAD
46	1	U1	CBTL04DP211	TFBGA48		NXP	CBTL04DP211	NXP	CBTL04DP211	LOAD
47	1	U2	PTN3460	HVQFN56		NXP	PTN3460BS	NXP	PTN3460BS	LOAD
48	1	U3	24C02	soic-8		STMicro	M24C02-WMN6TP	Digikey	497-8552-2-ND	LOAD
49	1	U4	LP3874	sot223-5		National	LP3874EMP-1.8/NOP8	Digikey	LP3874EMP-1.8TR-ND	LOAD